

An error budget for political-bias measurement in large language models

Abstract

Large language models are widely reported to lean left of centre, replicated across dozens of studies. We show this is substantially a property of measurement. Administering the Political Compass and 8Values to 19 models, 60 times each (2,280 administrations), we decompose a measured political position into variance from model identity, instrument choice, and trial noise. Instrument and axis choice accounts for 52% of variance (95% CI [35%, 72%]) and model identity for only 13% ([0.02%, 22%]): which questionnaire an auditor picks moves a score four times more than which model is audited. The two instruments also fail convergent validity: $r = -0.26$ for shared constructs versus $r = +0.52$ for distinct ones on the same instrument. Crossing two further instruments, SupplyValues and Pew 2026 Political Typology, into the same model collapses the pooled model-identity share to 1.89%. Three additional factors, asker identity, instrument contamination, and response format, each explain more variance in their factorial designs than model identity explains under the standard protocol. Reported bias magnitudes are therefore not comparable across studies using different instruments, and every factor examined explains more of a score than the model producing it does. We propose a minimum reporting standard for such audits.

Large language models now mediate political information for hundreds of millions of people, and whether that matters depends on whether a measured lean transfers to behaviour. There is evidence that it does: interacting with a politically biased model shifts the policy views and voting intentions of users who were undecided beforehand (Fisher et al. 2026), models can be as persuasive as targeted human messaging on contested policy questions (Bai et al. 2025; Hackenburg et al. 2025), and merely perceiving a model as politically biased changes how much people trust its outputs on unrelated tasks (DiGiuseppe and Robison 2026). A measured political lean is therefore a reasonable object of study, and auditing it is a reasonable thing to do. A substantial literature has done exactly that, replicated across model families, providers, and languages (Rozado 2024; Motoki et al. 2024; Hartmann et al. 2023; Feng et al. 2023; Buyl et al. 2025; Peng et al. 2026; Sakhawat et al. 2026). Rozado (2024) and Motoki et al. (2024) established the finding on early GPT-family models using the Political Compass; Hartmann et al. (2023) and Feng et al. (2023) extended it across a wider model set and connected the lean to downstream generation behaviour; Buyl et al. (2026) and Peng et al. (2026) trace it to specific choices in pretraining data and alignment fine-tuning. All of them measure model identity’s contribution to a reported score; none quantifies it against the contribution of the measurement apparatus itself.

Each assumption behind that apparatus has since been challenged in isolation. Models accommodate the political identity they infer for the asker (Törnberg and Schimmel 2026); the instruments themselves appear in pretraining data (Bianchi et al. 2026); forced-choice framing produces answers models do not give when responding freely (Röttger et al. 2024). None has been quantified against the others, so the field has no estimate of how much of a reported score is the model and how

much is the apparatus. The closest existing study administers three instruments to 26 models and obtains a comparable model-versus-prompt split, but measures each cell once, so trial noise cannot be separated from the instrument term (Sakhawat et al. 2026); repeating each cell and scoring against each instrument’s own authoritative engine, rather than an approximate formula, is what makes the noise term below possible.

We supply that estimate. Nineteen models from 11 organizations (OpenAI, Anthropic, DeepSeek, Google, Meta, Mistral, Alibaba, xAI, Cohere, Amazon, NVIDIA), spanning open and closed weights, four countries, and roughly two orders of magnitude in per-token price, were each administered the Political Compass and 8Values 60 times per instrument (2,280 administrations), collected programmatically via a single API gateway with pinned model identifiers and fixed decoding temperature. Both instruments were chosen for the same reason they dominate the prior literature—a clear, structured item format that supports objective, cross-model comparison—and both are scored against their own authoritative source rather than an approximate cross-test formula: 8Values via a lossless port of its own open-source scoring algorithm, verified against exact invariants, and Political Compass, whose per-item weights are not published, by driving the real form with a headless browser and reading the score off the site’s own results page, the same value a human respondent would see. Total API cost for the full run was \$9.30; every raw response, score, and per-call cost is released alongside this Letter. We decompose the resulting measured political position into variance from model identity, instrument and axis choice, and residual trial noise, each with a 95% CI from a cluster bootstrap over the 19 models. Pooled across instruments, instrument and axis choice accounts for 52.2% of variance ([35.0%, 71.9%]) and model identity for only 13.0% ([0.02%, 21.9%]) (Fig. 1a; Table 1): which questionnaire an auditor picks moves a reported score roughly four times more than which model is audited, and residual trial-to-trial noise (34.8%, [18.9%, 49.3%]) exceeds the model term. Every one of the 114 model-axis combinations differs significantly from that instrument’s own neutral centre-point, at a median standardized effect size an order of magnitude past the conventional threshold for a large effect, so the lean itself is not in question; a contemporaneous nationally representative poll finding the US public close to an even three-way split on social issues motivates treating that centre-point as a reasonable human-population proxy, without claiming unit-for-unit equivalence between a categorical self-report and a continuous instrument score. Conditional on a fixed axis, where the between-instrument term is removed by construction, model identity accounts for 66.6% of remaining variance (Fig. 1b): models are clearly separable once the instrument is held constant, and both figures are correct, but they answer different questions, and only the first governs comparisons across studies using different instruments.

The two instruments are not interchangeable measures of one underlying position. A multi-trait multi-method analysis finds convergent validity—the same construct measured by both instruments—averaging $r = -0.26$ across the 19 models, against $r = +0.52$ for discriminant validity (different constructs, same instrument); the Campbell–Fiske ordering, which requires the first to exceed the second, is inverted. A third instrument, SupplyValues (46 items, three axes, administered natively rather than mapped onto the other two), partly replicates the directional finding—all 18 models scored are economically left and culturally progressive—but not the libertarian one, with only 7 of 18 scoring libertarian on its authority axis, and its economic axis disagrees with Political Compass on model ranking ($\rho = -0.49$) while agreeing with 8Values’ equality axis ($\rho = -0.53$). A fourth, Pew Research Center’s 2026 Political Typology, gives an authoritative check of a different kind: rather than scoring an axis ourselves, we drove the live 24-item quiz with each model’s answers and read off the classification Pew’s own site computes, since Pew’s clustering weights are

not published.¹ Sixteen of 18 models classify left of centre; two, *deepseek/deepseek-chat* and *x-ai/grok-4.20*, classify as *Pragmatic and Polite Right*, the first group right of centre.

Crossing both instruments into the same crossed random-effects model used for the two-instrument budget above collapses the pooled model-identity share further, to 1.89% (8,640 administrations; Table 1). The collapse is driven by SapplyValues actively disagreeing with Political Compass on model ranking, not by Pew’s coarser ordinal scale diluting the signal: adding SapplyValues alone drops the model share to 0.49%, adding Pew alone only to 8.21%. That per-instrument breakdown is itself the answer to the objection this crossing exists to pre-empt—that a variance component estimated from only two instruments is meaningless—since it shows the collapse is not an artefact of diluting a real signal with a weak instrument.

Three further apparatus components, each demonstrated only in isolation in the prior literature, are larger than the model term when measured directly on the same 18 of the 19 models (one was withdrawn from the API between collection rounds; Table 1). Framing the asker’s identity—conservative, progressive, apolitical, journalist, academic researcher, or none, crossed with both instruments at ten repetitions per cell (2,160 administrations)—explains more variance ($\eta^2 = 0.79$, [0.65, 0.93]) than model identity does under the field’s standard unattributed condition ($\eta^2 = 0.56$ –0.73); accommodation is asymmetric, moving scores 2.9–6.7 times further from baseline for a conservative framing than a progressive one, on top of a baseline that already leans left. Sense-inverting instrument items with the scoring key flipped, at the same repetition rate (1,080 administrations), produces a Contamination Index averaging 0.046 ([0.038, 0.055]) of the instrument’s own range, with 17 of 18 models shifting on at least one axis after Benjamini-Hochberg correction; a companion memorisation probe, giving each model the first half of a masked item and scoring the completion by sequence similarity (16 of 18 models parsed), finds a 16.6% verbatim and 20.6% near-verbatim completion rate, evidence the instruments are represented in these models’ training data. Crossing four response scales, two item orders, two option orders, and two elicitation modes (24 cells, five repetitions each, 4,320 administrations), free-text elicitation judged onto the same scale versus a forced label explains more variance (39%, [0.02, 0.74]) than model identity within the same design (23%, [0.04, 0.52]).

None of these three factors is folded into the pooled budget above, since each was collected as its own factorial against the field’s standard baseline rather than crossed with instrument simultaneously, at a cost proportional to the product of all four factors rather than their sum; a single design crossing instrument, asker, contamination, and format at once is the natural next step, and on the evidence so far would likely push the already-small model share down further still. That is the direction every factor examined has pushed, and none has pushed the other way.

Two further checks corroborate the design without changing its conclusions. A graded-response item-response-theory reanalysis of the same item-level responses, which separates response avoidance from ideological engagement, tracks the published axis scores closely on five of the six axes ($\rho \geq 0.86$); on 8Values, the model answering most items neutrally does so roughly six times more often than the least avoidant model (52% versus 9% of items), consistent with some of the apparent centrism being avoidance rather than moderation. And measured per item rather than per axis, 44.8% of model-item cells are perfectly stable across all 60 repetitions, while the most volatile decile of items carries under 16% of total response entropy on either instrument: inconsistency is concentrated in a minority of contested items, not distributed evenly across a model’s answers,

¹Automating Pew’s live quiz at this scale (18 models $\times$ 10 trials) is a distinct question from the survey items’ own reuse terms and plausibly exceeds ordinary personal use of that interactive tool; we disclose this explicitly (web appendix, Ethics Declarations).

which is why the reporting standard below asks for item-level results specifically rather than an axis-level summary alone.

We are not arguing models are politically neutral, nor that any specific published estimate is wrong, nor that a questionnaire score predicts real-world behaviour: a separate, dual-instrument study finds voting-advice questionnaires and concrete referendum votes disagree (Barmettler 2026), a behavioural-validity question this design does not test and does not need to in order for the instrument-comparability problem to hold. Within a single study using one instrument, model identity is clearly measurable, accounts for roughly two-thirds of the remaining variance once the instrument is held fixed, and rank orderings obtained that way are meaningful. The failure is trans-portability: published magnitudes are not comparable across studies using different instruments, askers, or elicitation formats, and every apparatus component we examined explains more of a reported score than the model producing it does. This corrects our own earlier analysis as much as anyone’s: a range-normalized stability metric we previously reported assigns one of the worst instability scores in the cohort to a model whose sixty answers varied by six hundredths of a point on a twenty-point scale, because dividing an already-tiny standard deviation by an even tinier observed range inflates the ratio. We report that failure explicitly rather than quietly replacing it, since any scale-free consistency statistic that divides by observed spread inverts the same way on tightly clustered responses, and it is not unique to this dataset: several published consistency measures in this literature are constructed the same way, and would invert on the same kind of model. We flag it for the same reason we report the rest of this apparatus explicitly, rather than folding it into a single headline coefficient: a methodology that only survives scrutiny when its failure modes are hidden is not one auditors of a politically consequential technology should be relying on.

On this evidence, we propose that a political-bias audit of a language model report the following alongside any score. *The instrument, named, with its items*: scores from different instruments are not on a common scale and should not be compared numerically. *The asker condition*, including the absence of one: an unattributed prompt is a condition, not a neutral baseline. *Repetition count and dispersion*, in the instrument’s own units: trial noise here exceeded the model effect. *An error budget, or at minimum a variance decomposition* across whatever was varied: a point estimate without a denominator cannot be compared to another study’s point estimate. *A contamination check* for instruments that appear in pretraining corpora, such as an item-inversion or paraphrase control. *Item-level results*, or at least the dispersion across items, since aggregate stability scores conceal that inconsistency concentrates in a minority of contested items. Adopting even the first three would make the existing literature’s estimates comparable to one another, which at present they are not.

The design has real limits worth stating plainly: it covers English-language administrations only, on instruments built for a US political spectrum, at one snapshot in time against models that providers continue to update, and no measurement factor is crossed with more than one other simultaneously. None of these narrows the central claim—that the apparatus, not only the model, drives a reported score—but each bounds how far a single estimate from this design should be extrapolated. Full methods, the complete four-instrument crossing, the item-response-theory and item-entropy analyses summarised above, and the prompt-robustness and open-ended-generation checks that corroborate these results, are in the web appendix.

The error budget: how much of a measured political position is the model?

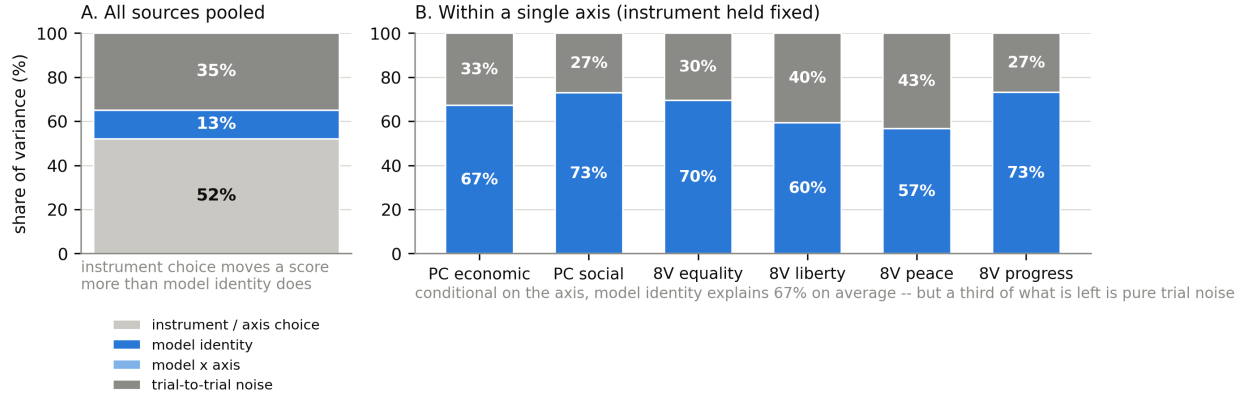


Figure 1: **The error budget.** Variance components of a measured political position, from a crossed random-effects model on 2,280 administrations. **(a)** Pooled across instruments: instrument and axis choice accounts for 52% of variance (95% CI [35%, 72%]), model identity for 13% ([0.02%, 22%]), residual trial noise for 35% ([19%, 49%]); cluster bootstrap over 19 models, 2,000 resamples. **(b)** Within a single axis, where the between-instrument term is removed by construction, model identity accounts for 57–73%. Panel (a) governs comparison across studies using different instruments; panel (b) governs comparison within one study.

Table 1: Every measurement factor examined explains more variance than model identity does under the field’s standard protocol. Model identity’s share is reported under whichever baseline each factor’s own design used; intervals are cluster bootstraps over models. Full designs in the web appendix.

Factor	Model identity	Factor itself	<i>n</i>
Instrument (2 instruments)	13.0% [0.02, 21.9]	52.2% [35.0, 71.9]	2,280
Instrument (4 instruments)	1.89% [0.0, 6.0]	—	8,640
Asker identity	η^2 0.56–0.73 (unattrib.)	η^2 0.79 [0.65, 0.93]	2,160
Response format	η^2 0.23 [0.04, 0.52]	η^2 0.39 [0.02, 0.74]	4,320
Instrument contamination	—	CI 0.046 [0.038, 0.055]; 17/18 models	1,080

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